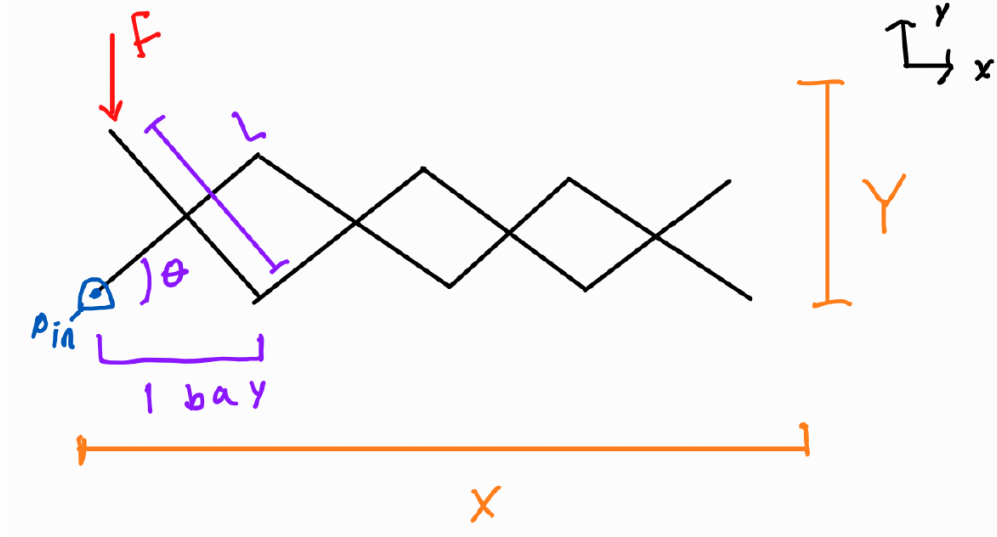




Assume a **quasi-static** system, i.e. motion is slow and inertia is neglected, so the actuator only overcomes friction.

Let θ = angle between each scissor link and the horizontal.

$$X(\theta) = nL \cos \theta, \quad X = \text{horizontal deployed length}$$

$$Y(\theta) = nL \sin \theta, \quad Y = \text{vertical actuator position}$$

$$\frac{dY}{d\theta} = nL \cos \theta, \quad \text{rate of vertical motion w.r.t. } \theta$$

Resistance consists of hinge and rail friction. With hinge torque τ_h and n_h hinges per bay,

$$\tau_{\text{hinges}} = nn_h \tau_h, \quad \tau_{\text{rail}} = F_{\text{rail}} \frac{dY}{d\theta}.$$

Thus,

$$\tau_{\theta} = \tau_{\text{hinges}} + \tau_{\text{rail}}.$$

Using virtual work¹,

$$F_{\text{act}} dY = \tau_{\theta} d\theta \Rightarrow F_{\text{act}} = \frac{\tau_{\theta}}{dY/d\theta}.$$

$$F_{\text{act}}(\theta) = \frac{nn_h \tau_h + F_{\text{rail}}(nL \cos \theta)}{nL \cos \theta}$$

This is the quasi-static actuator force required to move the scissor boom at angle θ .

¹Since Y and θ are in different coordinate systems, we must use virtual work and more precisely the principle of least action applied to energy in a static system.

Let α_k = absolute angle of link k , and $\theta_i = \alpha_{i+1} - \alpha_i$, where $\theta = 0$ is a fully straight joint.

$$\alpha_k = \alpha_k(t) + \sum_{j=1}^{k-1} \theta_j, \quad k = 2, \dots, n$$

$$X = \sum_{k=1}^n L \cos(\alpha_k), \quad X = \text{horizontal tip position}$$

$$Y = \sum_{k=1}^n L \sin(\alpha_k), \quad Y = \text{vertical tip position}$$

Initial folded configuration:

$$\alpha_k(0) = \frac{\pi}{2} + (-1)^{k-1} \phi_0, \quad \text{alternating link angles}$$

$$\theta_i(0) = \pm 2\phi_0, \quad \text{alternating joint angles}$$

Model each compliant joint as a circular arc:

$$R = \frac{s}{\theta}, \quad \mathbf{r} = \begin{bmatrix} R \sin \theta \\ R(1 - \cos \theta) \end{bmatrix} \quad \mathbf{F}_p = pA \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$$

Moment about the joint base:

$$M_p = \mathbf{r} \times \mathbf{F}_p \rightarrow \boxed{M_p = pA \frac{s}{\theta} (1 - \cos \theta)}$$

Joint resistance is modeled as a torsional spring:

$$\boxed{M_r = k\theta}$$

Quasi-static equilibrium:

$$M_p = M_r \rightarrow pA \frac{s}{\theta} (1 - \cos \theta) = k\theta$$

$$\boxed{p(\theta) = \frac{k\theta^2}{As(1 - \cos \theta)}}$$

This is the quasi-static pressure required to actuate each compliant joint as a function of θ .